ANDRES LEBBOS

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Heerlen, Netherlands

RESEARCH INTERESTS

Computational Oncology

Clinical Decision Systems

Stochastic Decision Making

Multi-Agent Systems

Network Dynamics

LANGUAGES

Spanish	*Native*	C2
English	*Professional*	C2
Arabic	*Advanced*	C1
Italian	*Moderate*	B1
Dutch	*Basic*	A2

PROFILE

AI researcher with a background in Computer Science and Electrical & Computer Engineering, currently pursuing a Master's in Artificial Intelligence. Focused on computational oncology and clinical decision support systems for treatment optimization and improved patient outcomes. Experienced in multi-agent systems and network-based modeling for complex systems.

EDUCATION

MSc Artificial Intelligence, Quantum Computing specialization

Maastricht University **Aug. 2024 - Sep. 2026 (expected)**

GPA **9.09 / 10** Maastricht, Netherlands

Thesis (*in progress*): Modeling angiogenesis and immunotherapy effects on simulated cancer tumor growth

Coursework: Advanced ML, Multi-Agent Systems, Dynamic Game Theory, Stochastic Decision Making, Quantum Algorithms

BSE Electrical & Computer Engineering, BSc Computer Science

Duke University **June 2016 - May 2019**

GPA **3.585 / 4** Durham, NC

RESEARCH PROJECTS

Computational Oncology

Master Thesis **Apr. 2026 - current**

Simulated tumor growth dynamics from microscopy videos using computer vision to extract temporal and topological properties.

Modeled tumor growth dynamics using network flow and swarm intelligence-based simulation of nutrient transport.

Investigated limitations of anti-angiogenic therapies and evaluated the effects of adaptive treatment strategies.

Multi-Agent Systems & Decision Theory

Swarm Intelligence for Search and Rescue **Nov. - Dec. 2025**

Developed a robot simulation using simultaneous localization and mapping (SLAM) for environment exploration and navigation.

Implemented particle swarm optimization based multi-agent coordination for efficient search and rescue tasks in robot swarms.

Strategic Voting with Uncertainty and Manipulation **Nov. 2025**

Analyzed the robustness of voting schemes under strategic vote manipulation and incomplete information, focusing on voter collusion and counter-strategic behavior.

TECH SKILLS

Programming Languages

Python	Java
MATLAB	C

Machine Learning & Data

PyTorch	TensorFlow
NumPy	SQL

Research Infrastructure

LaTeX	Unix
Hugging Face	GitHub

Software & Systems

AWS	Docker

AWARDS

Ibero-American Mathematical Olympiad

Bronze Medal 2013

Central American & Caribbean Mathematical Olympiad

Bronze Medal 2013

HOBBIES

3D Printing

Racket Sports

Board Games

Scuba Diving

Cooking

Fianco Game AI **Sep. - Oct. 2025**

Developed an AI agent for a chess-like game, achieving 7th place in a masters level competition against 70 other AI agents.

Multiple Robot Synchronized Path Planning **Aug. - Dec. 2018**

Designed an efficient, collision-free path-planning algorithm for dozens of synchronized robots in tight passages.

Quantum Algorithms & Stochastic Optimization

Security of Quartic-Speedup Algorithm **Sep. - Dec. 2025**

Analyzed the security implications of a quartic-speedup quantum algorithm, evaluating feasibility against classical baselines.

MCTS for Quantum Ansatz Design **Sep. - Dec. 2024**

Developed and evaluated Monte Carlo Tree Search (MCTS) variants for quantum ansatz design, improving ground energy estimation.

Robotics & Control Systems

Sonar Bathymetry Hardware **Jan. - May 2019**

Constructed an economic robot capable of diving up to 10,000 feet to track marine species and map oil reserves for research purposes.

Vehicle Slippage in Unknown Terrain **Aug. - Dec. 2018**

Created a learning algorithm for steering vehicles, reducing slippage through machine learning, feedback, and PID controllers.

WORK EXPERIENCE

Software Reliability Engineer **June 2022 - Feb. 2024**

Art of Problem Solving Bogota DC, Colombia

Maintained and modernized the reliability, resilience, infrastructure, and security of distributed systems serving over a million users.

Led a security committee, reducing system degrading incidents by 20% while improving their resolution and prevention.

Migrated and optimized a 1TB database, achieving 20% performance improvement and 40% cost reduction without any data loss.

Software Engineer **Sep. 2019 - June 2022**

Art of Problem Solving San Diego, CA

Designed and implemented software tools supporting an online learning platform with 10,000 students and 270,000 members.

Led candidate evaluation and hiring decisions, and mentored incoming engineers in complex systems and engineering workflows.

Developed an automated grading algorithm with 97% accuracy, including detection of academic dishonesty.